NVSHOP.LK E-Commerce Platform — Development & Implementation Report

Module: 2nd Year Software Engineering Project (GPSD)

Project Title: Development & Implementation of NVSHOP.LK E-Commerce Web System

Total Marks Allocation: 50 Marks

Target Organization: NVSHOP.LK (`185/1/2B New Road, Ambalangoda 80300, Sri Lanka`)

Document Version: 2.0 (Spring Boot 3.x Edition)

Document Summary & Marking Scheme Mapping

Section	Content Description	Page Allocation Guide	Marks
Section 1	**Software Development: Testing** (Justification of techniques, Test Case design, Test Logs)	~5 Pages	**20 Marks**
Section 2	**Implementation: Conversion and Training Plan** (Conversion strategy, Gantt Schedule, Resource Planning)	~4 Pages	**10 Marks**
Section 3	**Implementation: User Guide** (Installation, Operating procedures, Main functions, Error handling)	~10 Pages	**10 Marks**
Section 4	**Critical Appraisal (Combined Group & Individual)** (Macro perspective, Challenges, Future enhancements)	~5 Pages	**10 Marks**

SECTION 1: Software Development — Testing (20 Marks)

1.1 Justification of Testing Techniques (Tailored to NVSHOP.LK)

To ensure the reliability, accuracy, security, and responsiveness of the NVSHOP.LK e-commerce platform, a multi-layered testing strategy was designed and executed. Rather than applying generic software testing principles, each technique was strictly chosen to address specific operational and transactional risks inherent in retail computer hardware sales.

```
graph TD
    subgraph Testing Hierarchy for NVSHOP.LK
        EP_BVA["Equivalence Partitioning & Boundary Value Analysis\n(Validation of
Cart Quantities, LKR Prices & Forms)"]
        Integration["Integration Testing\n(React SPA <-> Java Spring Boot REST API &
Firebase Firestore)"]
        SystemTesting["System & Transactional End-to-End Testing\n(Order Placement ->
Admin Status Updates -> Inventory Sync)"]
        SecurityTesting["Security & Authentication Testing\n(Spring Security JWT
Tokens, Role-Based Access Control)"]
    end
    EP_BVA --> Integration
    Integration --> SystemTesting
    SystemTesting --> SecurityTesting
```

1. Equivalence Partitioning (EP)

- **Application in NVSHOP.LK:** Applied to input fields across the Storefront checkout form, VIP newsletter subscription, and Admin product creation tool. Inputs are partitioned into valid and invalid sets to minimize redundant test cases while maximizing coverage.
- **Justification:** For fields like phone numbers (`+94 XX XXX XXXX`), LKR prices (must be positive numeric values), and stock levels (non-negative integers), EP ensures that both standard user inputs and erroneous inputs are processed predictably without throwing uncaught runtime exceptions.

2. Boundary Value Analysis (BVA)

- **Application in NVSHOP.LK:** Concentrated on stock boundary conditions (e.g., purchasing 1 item when stock = 1 vs. attempting to purchase when stock = 0), cart item limit thresholds, and promo code inputs (e.g., `NVELITE1000`).
- **Justification:** In high-demand hardware retail (such as Nvidia RTX 4090 GPUs or ROG Strix laptops), stock overselling is a critical commercial failure. Testing boundary values ensures the system rejects invalid orders precisely when inventory reaches zero.

3. Integration Testing (Spring Boot & Cloud Synchronization)

- **Application in NVSHOP.LK:** Verifies data communication between the React 18 Frontend (`http://localhost:5173`), the Java Spring Boot REST API (`http://localhost:8080/api`), and Google Cloud Firestore (`orders` and `newsletter_subscribers` collections).
- **Justification:** NVSHOP.LK utilizes a dual-sync backend architecture. Integration testing guarantees that when a customer submits an order via `Checkout.jsx`, the POST request succeeds on Spring Boot (`OrderController`) while concurrently persisting to Cloud Firestore without blocking the UI thread.

4. System & End-to-End (E2E) Transaction Testing

- **Application in NVSHOP.LK:** Traces complete customer user journeys from initial product browsing $\rightarrow$ adding items to cart $\rightarrow$ newsletter voucher application $\rightarrow$ order placement $\rightarrow$ admin status fulfillment (`PROCESSING` $\rightarrow$ `SHIPPED` $\rightarrow$ `DELIVERED`).
- **Justification:** Validates that state changes in one module (e.g., updating order status in `Admin.jsx` via `AdminController`) immediately reflect across system metrics and inventory databases.

1.2 Choice & Justification of Test Cases

The test cases selected below represent critical functional vectors of the NVSHOP.LK system:

1. **Test Case TC-CART-01 (Cart Quantity Boundaries):** Verifies that cart quantities cannot be decremented below 1 or updated to negative values.
2. **Test Case TC-CHK-01 (Order Placement Dual Sync):** Validates that submitting an order writes valid transactional payloads to both Spring Boot MySQL DB and Cloud Firestore.
3. **Test Case TC-VIP-01 (Newsletter Voucher Generation):** Validates email format enforcement and instant generation of voucher `NVELITE1000`.
4. **Test Case TC-ADM-01 (Product Price & Stock Update):** Validates admin CRUD inventory updates with immediate storefront price rendering.
5. **Test Case TC-AUTH-01 (Admin Dashboard Access Control):** Ensures unauthenticated or non-admin customer roles are restricted from accessing `/admin` endpoints.

1.3 Test Logs (Execution Results)

Test Case ID	Test Case Description & Target Module	Input Data Values	Expected Result	Actual Result	Status	Action Taken / Debugging
TC-	Cart	Product: *Asus ROG Strix*	Quantity remains	Quantity	**PASS**	None

CART-01	decrement boundary test (`Cart.jsx`)	*G16* Current Qty: 1 Click - button	1; remove option prompted	stayed at 1 with toast notification		required.
TC-CART-02	Cart item addition & total calculation	Add *RTX 4090* (LKR 750,000) x 2 + *Curved Monitor* (LKR 120,000) x 1	Subtotal = LKR 1,620,000	Calculated LKR 1,620,000 correctly	**PASS**	Verified standard LKR currency formatter.
TC-VIP-01	Newsletter valid email submission (`Home.jsx`)	Email: `customer@gmail.com`	Success toast, Firestore document added, Voucher `NVELITE1000` rendered	Voucher displayed, Firestore updated	**PASS**	Verified async Firestore write completion.
TC-VIP-02	Newsletter invalid email format (`Home.jsx`)	Email: `invalid-email-text`	Inline validation error: "Please enter a valid email address"	Error message displayed, form blocked	**PASS**	Native regex check operational.
TC-CHK-01	Complete checkout order submission (`Checkout.jsx`)	Name: "Kamal Perera" Phone: `0771234567` Address: "Galle Road, Colombo" Payment: "COD"	POST to `/api/orders` returns HTTP 201; Firestore `orders` doc created; Cart cleared	HTTP 201 received; Firestore updated; redirected to order confirmation	**PASS**	Dual-sync pipeline verified.
TC-ADM-01	Admin inventory stock update (`Admin.jsx`)	Product ID: 5 (*Intel i9-14900K*) New Stock: `15` New Price: `LKR 215,000`	PUT request to `/api/products/5` succeeds; Admin table & Storefront update	Values updated instantly across DB and UI	**PASS**	State refreshed in React context.
TC-ADM-02	Admin order status transition (`Admin.jsx`)	Order ID: `#NV-1002` Action: Change status from `PROCESSING` to `SHIPPED`	PATCH request updates DB status; Status pill turns blue (`SHIPPED`)	Status updated instantly to `SHIPPED`	**PASS**	State sync verified.

TC-AUTH-01	Unauthorized Admin Access attempt (`Admin.jsx`)	Role: `Customer` Target URL: `/admin`	Redirected to Login / Access Denied banner shown	Redirected to login modal with notification	**PASS**	Role-based router guard enforced.

SECTION 2: Implementation — Conversion and Training Plan (10 Marks)

2.1 Justification of Conversion Technique

For NVSHOP.LK, transitioning from the traditional physical showroom operation in Ambalangoda to the automated full-stack e-commerce web platform requires a structured strategy to ensure continuous retail business operations without revenue loss.

Chosen Technique: **Phased Conversion (With Parallel Pilot Verification)**

- **Direct Cutover (Rejected):** High risk of operational failure. If online database services experience unexpected downtime, physical sales processing could stall.
- **Parallel Conversion (Stage 1):** Runs both manual sales recording and the new digital platform side-by-side for 2 weeks to verify stock sync accuracy.
- **Phased Conversion (Primary Approach):**
- **Phase A (Internal Administration):** Roll out the Admin Dashboard (`Admin.jsx`) to inventory managers to digitize product catalog and stock levels (48 items).
- **Phase B (VIP & Local Pilot):** Launch "NV Elite" VIP lead registration and local Ambalangoda click-and-collect orders.
- **Phase C (Island-wide Public Rollout):** Enable full island-wide shipping, online payments, and automated order status tracking across Sri Lanka.

2.2 Schedule & Planning for Conversion

Implementation & Deployment Schedule

```
Task Name                  W1  W2  W3  W4  W5  W6  W7  W8
-------------------------------------------------------------
Environment & Server Setup  [XX]
Database & API Seeding          [XX]
```

```
Admin System Pilot & Training        [XX]
Parallel Stock Logging Setup             [XX]
Customer Storefront Beta Testing             [XX]
Full System Cutover & Go-Live                    [XX]
Post-Implementation Audit                            [XX]
-------------------------------------------------------------------
```

Narrative Resource Allocation (Who, What, Where, When, How)

6. **Who (Stakeholders & Team Roles):**

- **Lead Software Engineer:** Oversees server deployment, API routes (`Spring Boot 3.x`), and Firebase setup.
- **Store Manager (Ambalangoda Showroom):** Validates stock accuracy, product descriptions, and LKR pricing.
- **Sales & Customer Support Staff:** Trained on operating the Admin Dashboard (`Admin.jsx`), order status fulfillment, and customer inquiries.

7. **What (Infrastructure & Tooling):**

- Node.js (v18+) build environment, Java 17/21 SDK, Gradle wrapper (`./gradlew`), MySQL Server instance (`gpsd_project`), Google Cloud Console (Firebase Firestore project `nvshop-lk`).

8. **Where (Locations):**

- **Physical Site:** NVSHOP Showroom (`185/1/2B New Road, Ambalangoda 80300`).
- **Digital Site:** Cloud Hosting / Production Web Server (`http://localhost:5173` local / production domain).

9. **When (Execution Timeline):**

- 6-Week phased deployment timeline with post-implementation support in Weeks 7–8.

10. **How (Training & Operational Procedures):**

- Conduct 3 hands-on staff training workshops on managing products and processing orders.
- Provide printed and digital versions of the **NVSHOP User Guide**.

SECTION 3: Implementation — User Guide (10 Marks)

3.1 Installation & System Setup Instructions

Environment Prerequisites

- **Node.js** (v18.0.0 or higher)

- **Java Development Kit (JDK)** (v17 or v21)
- **Gradle** (Included via `gradlew` wrapper)
- **MySQL Database Server** (Running on port 3306 with database `gpsd_project`)
- **Modern Web Browser** (Google Chrome, Microsoft Edge, Brave, or Firefox)

Step-by-Step Installation Commands

1. Setting Up the Java Spring Boot 3.x Backend API

```
# Navigate to the Spring Boot backend directory
cd "Back-end/Back_End"

# Run Gradle build and launch Spring Boot Server on Port 8080
./gradlew bootRun
```

Expected Server Output: `Started BackEndApplication in X.XXX seconds (process running on port 8080)`

2. Setting Up the React Frontend Web Application

```
# Navigate to the React frontend directory
cd "Front-end/frontend"

# Install Node modules dependencies
npm install

# Launch Vite Development Server
npm run dev
```

Expected Server Output: `VITE v5.x.x ready in XXX ms` $\rightarrow$ `Local: http://localhost:5173/`

3.2 Operating Instructions & Main System Functions

Function 1: Browsing & Filtering Hardware Inventory

11. Open your browser and navigate to `http://localhost:5173`.
12. Use the top navigation menu or category pills to filter products by 11 main hardware categories (e.g., *Laptops*, *Graphic Cards*, *Processors*, *Monitors*).
13. Use the global **Search Bar** to locate specific components by name or model (e.g., "RTX 4090", "i9-14900K").

Function 2: Managing Shopping Cart & Placing Orders

14. Click **Add to Cart** on any desired product card.
15. Open the slide-over **Shopping Cart** to inspect selected items, adjust quantities using `+` / `-` buttons, or remove items.
16. Click **Proceed to Checkout** to open the order placement modal.
17. Fill in required customer details:

- Full Name, Contact Phone Number, Email Address, Delivery Address, and City.
- Select Preferred Payment Method (Cash on Delivery, Bank Transfer, Online Card Payment).

18. Click **Confirm & Place Order**. The system will save the order to Spring Boot REST API (`OrderController.java`) and Cloud Firestore, outputting an instant confirmation receipt.

Function 3: Joining "NV Elite" VIP Loyalty Program

19. Scroll to the bottom newsletter section on the homepage ("Join the NV Elite").
20. Enter your email address and click **Join Now**.
21. Upon validation, the system generates your unique **LKR 1,000 Discount Voucher** (`NVELITE1000`).
22. Click **Copy Code** to copy the promo code directly to your clipboard.

Function 4: Admin Dashboard & Inventory Management

23. Click **Admin** in the top navigation header.
24. Sign in with administrator credentials:

- **Email:** `admin@example.com`
- **Password:** `admin123`

25. **Managing Orders:** View incoming orders, click on status tags to update from `PROCESSING` to `SHIPPED` or `DELIVERED`.
26. **Managing Products:**

- Click **Add Product** to list new hardware items with image URL, price, stock count, and description.
- Click **Edit** on existing rows to adjust LKR prices or replenish stock levels.

3.3 General Error-Handling & Troubleshooting Guide

Error Condition / Message	Root Cause	Resolution / Recovery Action
"Failed to fetch products from server"	Spring Boot API server is not running on port 8080 or MySQL service is stopped.	Start MySQL and run `./gradlew bootRun` in `Back-end/Back_End`.
"Please enter a valid email address"	Invalid email format submitted in NV Elite signup form.	Ensure email includes `@` and a valid domain (e.g., `user@domain.com`).

"Out of Stock" warning on button	Selected product inventory count equals 0.	Restock inventory via Admin Dashboard (`Admin.jsx`).
"Access Denied / Admin Privileges Required"	Non-admin user attempted to load `/admin` route.	Login using `admin@example.com` account.
Firebase Firestore sync delay	Temporary network interruption or missing API keys.	Verify internet connection; check MySQL database logs.

SECTION 4: Critical Appraisal (10 Marks)

4.1 Combined Group Critical Appraisal (5 Marks)

Macro-Perspective of Completed System

The development of NVSHOP.LK successfully delivers a high-performance e-commerce ecosystem tailored to Sri Lanka's tech retail market. Combining a lightweight React single-page frontend with Java Spring Boot 3.x and Google Cloud Firestore established a robust, resilient 3-tier system architecture capable of serving high traffic during flash sales and product launches.

Key Technical Challenges Faced

27. **Dual-Sync Asynchronous State Management:** Synchronizing order states simultaneously between a Spring Data JPA relational database (MySQL `gpsd_project`) and an asynchronous Cloud NoSQL database (Firestore) caused race conditions in early development builds. This was resolved by implementing clean async-await wrappers and fallback handlers in `src/services/api.js`.
28. **Dynamic LKR Currency & Precision Handling:** Standardizing numeric operations across product catalogs, shipping fees, discount vouchers, and admin revenue statistics required establishing centralized utility functions (`formatLKR`) to prevent float rounding errors.

Future System Enhancements Roadmap

29. **Live Payment Gateway Integration:** Incorporate PayHere / IPG integration for instant online credit card processing in LKR.
30. **Automated Customer Email/SMS Receipts:** Integrate Twilio SMS API and Spring Boot JavaMailSender to send automated SMS dispatch notifications to Sri Lankan customer phone numbers.
31. **Progressive Web App (PWA) Mobile Support:** Enable offline catalog caching and push notifications for mobile users.

4.2 Individual Critical Appraisal (5 Marks)

Member Breakdown & Module Ownership

Member 1: Frontend Architecture & User Experience Lead

- **Assigned Responsibilities:** Design system implementation (Tailwind CSS, dark mode aesthetics), storefront page components (`Home.jsx`, `Catalog.jsx`, `Cart.jsx`, `Checkout.jsx`), hero banners, and NV Elite VIP loyalty feature.
- **Critical Assessment:** The UI achieves high visual fidelity and mobile responsiveness. Micro-animations and toast notifications provide clear interactive feedback. Future work should focus on accessibility (ARIA labels) and localized language support (Sinhala / Tamil translations).

Member 2: Backend API & Cloud Database Lead

- **Assigned Responsibilities:** Java Spring Boot REST API controllers (`OrderController.java`, `ProductController.java`, `AuthController.java`), JPA entity mappings, MySQL database schemas, and Firebase Firestore initialization.
- **Critical Assessment:** Restful standards were strictly maintained across all endpoints. The dual-sync strategy ensures cloud redundancy. Implementing Spring Security JWT token authentication provided robust protection against unauthorized REST calls.

Member 3: Admin Dashboard & Quality Assurance Lead

- **Assigned Responsibilities:** Admin metrics dashboard (`Admin.jsx`), order status management, inventory CRUD operations, system testing execution, and test log documentation.
- **Critical Assessment:** The admin portal empowers non-technical showroom staff in Ambalangoda to manage business operations efficiently. Real-time metric updates and low-stock alerts significantly reduce manual inventory inspection time.

Appendix: Official Business & System Metadata

- **Business Name:** NVSHOP.LK
- **Physical Showroom Address:** `185/1/2B New Road, Ambalangoda 80300, Sri Lanka`
- **Contact Hotlines:** `+94 76 989 0079` / `+94 77 747 0186`
- **Frontend Port:** `http://localhost:5173`
- **Spring Boot API Port:** `http://localhost:8080/api`